

An intelligent DDoS attack detection tree-based model using Gini index feature selection method

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ABSTRACT

Cyber security has recently garnered enormous attention due to the popularity of the Internet of Things (IoT), intelligent devices' rapid growth, and a vast number of real-life applications. As a result, detecting threats and constructing an efficient Intrusion detection system (IDS) have become crucial in today's security requirements. Withal, the large amount of high dimensional data might influence detection effectiveness and raise the computation requirements. Artificial Intelligence (AI) has recently attracted much attention and is widely used to build intelligent IDSs to preserve data confidentiality, integrity, and availability. Distributed denial of service (DDoS) is a denial of service (DoS) variant mainly targeting asset availability. Preventing DoS at the network or infrastructure level typically depends on implementing an IDS. This paper proposes a novel intelligent DDoS attack detection model based on a Decision Tree (DT) algorithm and an enhanced Gini index feature selection method. Our approach is evaluated on the UNSW-NB15 dataset, which contains 1,140,045 samples and is more recent and comprehensive than those used in previous works. Our system achieved an overall accuracy of 98%, outperforming baseline models that used more advanced algorithms such as Random Forest and XGBoost. Our enhanced Gini index feature selection method allowed us to select only 13 out of 45 security features, significantly reducing the data dimensionality and avoiding overfitting issues. Our model also has a lower false alarm rate, misclassifying only 2% of the testing instances. Our approach is, therefore, highly effective and efficient, with the potential to be used in real-world network security applications.

1. Introduction

The need for cybersecurity and protection against cyber-attacks is continuously growing, propelled by various factors, including the widespread adoption of the IoT, rapid expansion of computer networks, and a diverse range of related applications utilized by individuals and companies for personal or commercial purposes. Cyber-attacks like DDoS, computer viruses, and unauthorised access can cause irreversible harm and significant financial losses for large-scale networks [1,2]. Cyber-attacks' impact is not limited to specific industries but affects various sectors. A ransomware outbreak in May 2017 is a prime example of this. The attack resulted in a staggering \$8 billion worth of losses for companies and institutions in several areas, such as finance, healthcare, energy, and education [3]. One of the most significant cybersecurity issues is the DDoS attack, which aims to deprive a resource of

availability or reduce its service quality for legitimate users. A DDoS attack can be launched using diverse methods and target various OSI and TCP/IP model layers. The type of software tool used to generate the attack traffic, the communication layer, the victim's type, and the targeted protocol are all critical factors that affect how a DDoS attack is executed [4]. Diverse approaches are employed in DDoS attacks to exploit various weaknesses and target different resources, as presented in Table 1 [1]. Attackers' pursuit of high processing power has resulted in the emergence of DDoS attacks. DDoS attacks significantly increase the potency of a DDoS attack when attackers generate a vast volume of traffic in a brief period. To execute a successful DDoS attack, attackers must identify vulnerabilities that can be exploited. Different techniques include flooding the network or resource with excessive traffic, exploiting software vulnerabilities, or leveraging botnets to launch attacks. Therefore, organizations must develop countermeasures to

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Table 1
Common DDoS attack types [4] Types.

Type	Target	Vulnerability
Flooding Devices	Network or server. Network hardware components.	Resource limitations. Bugs in firmware/software
Protocol	Protocol services	IP spoofing and TCP SYN flooding are examples of protocol flaws.
Application Operating systems	Application services OS services	Limitations and flaws in the application OS software components bugs.

mitigate the risk of DDoS attacks and implement adequate cybersecurity measures to ensure business continuity.

IoT and its real-world applications are a hot research topic nowadays. The characteristics of IoT make it easier to adapt to real-world applications on one side, but they also expose it to cyber threats. A DDoS attack is one of the most damaging attacks, which prevents legitimate users from accessing services they have paid for [5,6]. This breaches the Service Level Agreement (SLA) conditions, resulting in significant financial losses for businesses and organizations. Furthermore, DDoS impacts services provided by small networks, such as smart homes, healthcare, and agriculture [7]. DoS attacks on essential innovative applications like healthcare may even result in death due to delays in routine services. IoT devices (such as bright light bulbs, smart door locks, and smart televisions) are a popular target for attackers who use their flaws to launch DDoS attacks [6,8–10]. As a result, one of the most pressing concerns for researchers nowadays is the security of these devices [11,12]. Researchers worldwide have extensively studied intrusion detection mechanisms to mitigate security risks [13]. Intrusion detection systems (IDSs) are broadly divided into two types: anomaly-based and signature-based [14,15]. A signature-based IDS compares network traffic patterns to previously-stored attack patterns (signatures). If a match is identified, an alarm is triggered. Although signature-based IDSs have a high level of accuracy and a low number of false alarms, they cannot detect unknown attack patterns [16]. In contrast, an anomaly-based IDS continuously monitors network traffic for deviations from the normal network profile, raising the alarm if a deviation exceeds the threshold, indicating the detection of an attack [17]. This paper focuses on the anomaly-based IDS approach.

Machine learning (ML) methods are used to learn the normal network profile. Anomaly-based IDSs are favored over signature-based IDSs since they can detect new attack patterns, although this increased sensitivity comes with a higher rate of false alarms [18]. False alarms occur when an IDS misclassifies a regular event as an attack, while a false positive detection is when an IDS classifies an attack as a recurring event. Intrusion detection is a binary classification task in which a trained classifier seeks to categorize network traffic into normal or attack classes with the highest accuracy and the lowest false alarms possible. Once trained, the detection model can effectively identify and classify new attacks with high accuracy [19].

Meanwhile, the effectiveness and superiority of detection methods have been a critical focus for researchers. In this regard, this paper aims to propose an intelligent DDoS attack detection model based on a DT, which improves detection accuracy while reducing computation complexity. Furthermore, the effectiveness and superiority of the proposed method are verified clearly by the experimental results.

The following points justify the novelty and contribution of this research:

- 1 We introduce a DDoS attack detection model based on the concept of DT to identify and forecast DDoS attacks in cyberspace efficiently.
- 2 To determine the most significant features, we propose a method for selecting and ranking security features, which the suggested intelligent tree-based DDoS prediction model addresses.

- 3 Compared to baseline work, the proposed model is implemented with a larger dataset of 1140,045 records for IDS. We evaluate the proposed model using established performance evaluation criteria, such as accuracy, Precision, Recall, and Fscore.

The rest of the paper is organized as follows. Section 2 looks into similar work on DDoS attack detection models. Section 3 describes and discusses our intelligent tree-based DDoS detection model. Our experiments and results discussion of the suggested model is presented in Section 4. The final section wraps up the paper and offers the conclusion of this work.

2. Literature review

DDoS attacks pose a significant threat to network security, and traditional defense mechanisms are often insufficient to detect and mitigate them. In recent years, ML approaches have emerged as promising solutions for detecting and classifying DDoS attacks. This literature review examines several ML-based DDoS detection approaches and evaluates their strengths and weaknesses. Our review includes neural network-based models, ensemble classification methods, statistical learning-based botnet identification tools, and other novel approaches. We also discuss the importance of data preprocessing and feature engineering in improving the performance of ML models.

A comparative analysis of port scanning and DDoS attacks is presented by Aamir et al. [19] using the recently published benchmark dataset CICIDS2017. Twenty-two different ML algorithms have been implemented and evaluated to see how well they perform against the most recent attack vectors. According to the classification results, the Support Vector Machine (SVM) yields more than 90% accuracy. Can et al. [20] introduce a neural network-based model to detect and classify DDoS attacks. The model takes advantage of a proposed automatic FS approach, and the model's effectiveness was assessed using the CICDDoS2019 dataset and demonstrated the effectiveness of the proposed model. However, the model is significantly affected by the discrepancy between the data distribution in training and testing data. Finally, Aamir and Ali Zaidi presented two methods [21,22] to enhance the effectiveness of IDSs. In [21], a strategic-level framework integrates feature engineering and machine learning into a specified experimentation flow, validated using cross-validation and area-under-curve evaluations. In [22], the study employed a semi-supervised ML technique to classify DDoS attacks, achieving high accuracy scores using CNN, SVM, and RF models with reduced feature space. These studies highlight the importance of incorporating advanced machine learning techniques into IDSs to improve performance and identify cyber threats more accurately.

In [23], Rambabu and Venkatram propose an ensemble classification approach that uses traffic flow measurements to protect IoT networks from DDoS attacks. The training corpus is partitioned into multiple clusters, and each cluster's corresponding traffic flow metric values are identified and utilized as features. Experimental results demonstrate that the suggested method outperforms contemporary methods in detecting DDoS attacks while generating fewer false alarms. Novaes et al. [24] introduced a security model based on adversarial training in Software-Defined Networking (SDN). The system uses the Generative Adversary Network (GAN) method to detect DDoS attempts and relies on adversarial training to make the system less sensitive to adversarial attacks. Martínez Garre et al. [25] used the WEKA tool and the NSL-KDD dataset to implement a ping of death detection model based on the RF algorithm. The model was able to achieve 99.76% of accuracy.

Jayalaxmi et al. [26] developed a feature extraction model using the amalgamation of ML and Deep Learning (DL) techniques for the Industrial Internet of Things (IIoT). ML algorithms and DL feature extraction methods evaluate the most compelling feature set. The model is trained with the most persuasive feature set evaluated using ML algorithms and DL feature extraction methods. Gadallah et al. [27] created an ML-based technique to predict a DDoS threat on the SDN architecture. The

developed model uses unconventional features such as type of service header, transport layer protocol header, packet inter-arrival time, and unknown destination addresses. The developed model is trained with kernel radial basis function. Using the K-Nearest Neighbor (KNN), RF, Gradient Boosting, Adaboost, DT, and Linear Discriminant Analysis methods, Karatas et al. [28] propose six machine learning-based IDS that employ various methods and the current security dataset (CSE-CI-C-IDS2018). A synthetic data generation model (SMOTE) is utilized to improve the detection rate of infrequent incursions. The proposed approach significantly boosts the detection rate for rare attacks and increases accuracy by 4.01% to 30.59%.

Kshirsagar and Kumar [29] present an ensemble of filter feature selection techniques (EFFST) to generate a significant feature subset for web assault detection and improve IDS efficiency. The proposed scheme has a high detection rate of 99.9909% using J48 and 24 features on the CICIDS 2017 dataset. Agarwal et al. [30] used the Whale optimization algorithm and neural network to propose a novel DDoS detection scheme. The suggested technique was evaluated in this study and simulated using the MATLAB tool, resulting in a 95.35% detection of DDoS attacks. Ahuja et al. [31] suggested a hybrid Support Vector Classifier (SVC) and RF. The model classifies traffic using SVC before being filtered by RF again, with an accuracy of 98.8%. The models are trained using the SDN dataset generated using the MININET emulator.

Akgun et al. [32] propose an IDS for identifying DDoS attacks using a deep learning model. The study evaluates various models based on Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), and Long Short Term Memory (LSTM) and found that the CNN-based inception model provides the best results with 99.99% accuracy for binary and 99.30% for multiclass classification. Characterization et al. [33] presented a novel comparison using multiple datasets, such as UGR16 and UNSW-NB15. The study's findings show that the data preparation tasks are crucial for the ML algorithm to classify anomalies better despite the dataset quality. This study's most relevant conclusion is the importance of preprocessing stages, such as statistical feature scaling and important feature selection. Tama & Lim [34] conducted a systematic mapping study and a comparative analysis of ensemble learning for IDS. This work uses thorough mapping research to provide an overview of how ensemble learners are used in IDSs. They gathered and examined 124 notable papers. Years of publication, publication locations, datasets, ensemble methodologies, and IDS techniques were all used to categorize the selected works.

Moreover, Ashraf et al. [14] presented IoTBoT-IDS, a novel statistical learning-based botnet identification tool for protecting intelligent city networks. The botnet uses Beta Mixture Model (BMM) parameters and Correntropy models to detect attacks launched by popular IoT devices. Bouke et al. [35] present an Enhanced Intelligent Intrusion Detection Model (E2IDS) to detect cyberattacks. The model is based on the DT algorithm and uses data balancing and feature selection approaches to address the complexity and volume of data in IoT networks. The proposed feature selection approach reduces the dimensionality of the data from 41 to 11 features and achieves an accuracy score of 98.8%. Roopak et al. [36,37] proposed two methods for enhancing the security of IoT networks against cyber-attacks. The first study suggested a feature selection method for detecting DDoS attacks using a multi-objective optimization approach. The proposed method achieved high accuracy (99.9%) and reduced the total number of features by nearly 90%. The second study suggested using DL models for cyber security in IoT networks, specifically for detecting DDoS attacks. The hybrid CNN+LSTM model performed the best, achieving an accuracy of 97.16%. Daneshgadeh Çakmakçı et al. [38] proposed a DDoS detection system for multidimensional data that is continuous and sequential. The proposed method utilizes a chi-square test, the Mahalanobis distance, and a kernel-based learning algorithm. Obeidat and Yaqbeh [39] Propose a machine learning-based method to detect botnet activities, which are a common threat on the internet for illegal activities such as click fraud, DDOS attacks, and spam production. The proposed method uses genetic

algorithm-based classifiers to identify botnet behavior in data traffic by categorizing it based on time slots. The experimental results using existing datasets show that the proposed method can detect botnet activities with High Precision. Adnan et al. [40] introduce a novel approach for intrusion detection in IoT-based wireless sensor networks. The proposed Intelligent IDS system combines a rule-based feature selection algorithm with a Multi-Objective optimization (PSO) technique and a multiclass SVM classification algorithm with an enhanced rule-based approach. The system's performance was evaluated using the KDD'99 Cup and CIDD datasets. The results showed that the proposed IDS system improved detection accuracy and reduced the false positive rate.

Kumari and Jain [41] thoroughly analyze DDoS attacks over the Internet of Things (IoT) and the existing defense methodologies. The authors explore various DDoS variants and IoT security issues, including exploiting IoT devices and creating botnets. They also compare and analyze defense mechanisms, including machine learning and software-defined networking (SDN). The paper concludes that more intelligent and smarter solutions are needed to combat the new variants of DDoS attacks and attacking methods. Next, Priyadarshini and Barik [42] propose a new mechanism for defending against DDoS attacks in fog computing environments using Software Defined Network (SDN) technology and deep learning. The mechanism uses SDN to deploy a DDoS defense module to detect anomalous behavior of DDoS attacks in network/transport levels. The proposed model was trained using historical data and tested with both simulated and real DDoS attack packets, achieving an accuracy of 98.88. Finally, Ismail et al. [43] proposed a framework for DDoS attack prediction. The proposed work used RF and XGBoost classification algorithms, and experiments conducted using the UNSW-NB15 dataset showed that the average model accuracy is 90%.

To this end, based on the reviewed literature, it is clear that machine learning approaches are increasingly being used to detect and classify Distributed Denial-of-Service (DDoS) attacks. The studies reviewed include neural network-based models, ensemble classification methods, statistical learning-based botnet identification tools, and other novel approaches. Additionally, data preprocessing and feature engineering were important factors in improving the performance of the ML models.

However, a common gap in many of the studies reviewed is that they relied on small datasets, which may not reflect the real-world scenario. To address this gap, our approach uses a massive dataset of more than one million records to improve the accuracy and reliability of the models. Furthermore, we use an enhanced Gini index feature selection approach to reduce the data dimensionality and avoid overfitting issues.

3. Research methods

The proposed intelligent DDoS attack detection system, based on the DT and Gini Index (GI) FS method, is intended to automatically detect potential DDoS attacks and security policy violations and alert administrators. As described in the previous section, DDoS attacks occur when multiple machines send fictitious attack packets to the victim's network elements, potentially resulting in interruptions and service delays, and the theft of critical corporate data. The architecture of the proposed system is illustrated in Fig. 1. To develop this system, we followed a research methodology that involved several steps, which are outlined in the research method section.

3.1. Dataset

In any ML project, the first stage in the implementation process is to prepare the dataset. According to [44–48], most machine learning algorithms require data to be explicitly formatted, so datasets generally require some preparation before yielding valuable insights. Analysts can trust, understand, and ask better questions about their data, so thorough data preparation makes their analyses more accurate and insightful.

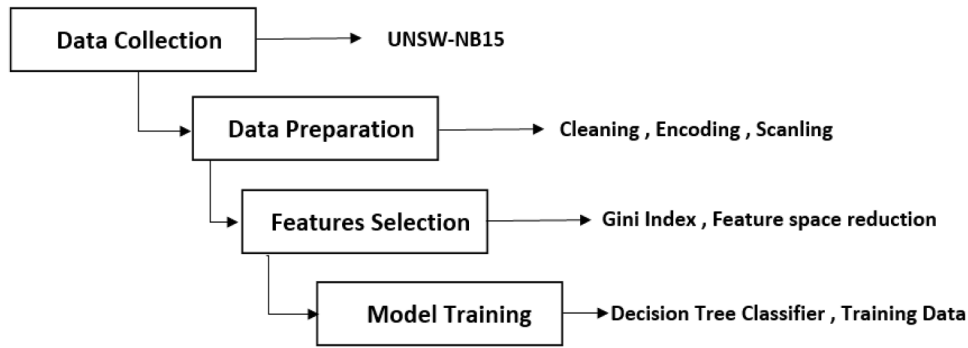


Fig. 1. Overall architecture of the proposed model.

Improved meaningful data analysis leads to better findings and better results. According to [49], CIDD5-001, ADFA, DARPA, KDDCUP 99, KDD98, CIDD5-001, and NSLKDD are some older datasets that IDS researchers have used. Despite this, these datasets (most developed more than 20 years ago) have several flaws that make them untrustworthy and obsolete. These datasets are no longer realistic representations of modern threat environments, and models built on them will not produce reliable results. Due to these limitations, the UNSW-NB15¹ dataset was generated by the Australian Center for Cyber Security (ACCS) in partnership with several researchers worldwide. Using the IXIA PerfectStorm tool, they developed a rich hybrid collection of normal and abnormal modern network traffic [50]. As indicated in Table 2, the total number of entries saved in the four Comma-separated values (CSV) files is 2540,047. In our experiments, we used UNSW-NB15.3 and UNSW-NB15.4 to merge these two files into one CSV file containing 1140,045 samples. 70% of these instances were used in training, and 30% were used for testing. The dataset includes 49 properties that determine the characteristics of each record. As seen in Table 3, the features type varies, with some being nominal, numeric, and others taking on time-stamp values [51]. The nominal features are *proto*, *service*, *state*, and *attack_cat* (colored in blue see Table 3). Out of 49 security features, only 41 are relevant to our work. We will exclude the *Label* class since it is out of the scope of the paper. Also, *srcip*, *sport*, *dstip*, *dsport*, *Stime*, and *Ltime* will be dropped as suggested by the dataset creators in [50].

The UNSW-NB15 dataset includes the most recent higher attacks, mimicking current network conditions. Table 4 lists the ten different categories of traffic in the dataset: *Normal*, *Fuzzing*, *Analysis*, *Backdoor*, *DOS*, *Exploits*, *Generic*, *Reconnaissance*, and *Worms*.

3.2. Data preparation

DP (also known as “data preprocessing”) techniques refer to the modification, deletion, or changing of data from a training set [44]. DP can make or break a model’s predictive ability. The default DP steps for any ML project are Encoding, Scaling, and FS. According to [44,52],

Table 2
The dataset CSV files.

File Name	Extension	Number of records
UNSW-NB15_1	CSV	700,001
UNSW-NB15_2	CSV	700,001
UNSW-NB15_3	CSV	700,001
UNSW-NB15_4	CSV	700,001
		2540,047
		1140,045 records

Table 3
Security features of the UNSW-NB 15 datasets.

No.	Name	Type	No.	Name	Type
1	srcip	nominal	26	res_bdy_len	integer
2	sport	integer	27	Sjit	Float
3	dstip	nominal	28	Djit	Float
4	dsport	integer	29	Stime	Time-stamp
5	proto	nominal	30	Ltime	Time-stamp
6	state	nominal	31	Sintpkt	Float
7	dur	Float	32	Dintpkt	Float
8	sbytes	Integer	33	tcprtt	Float
9	dbytes	Integer	34	synack	Float
10	sttl	Integer	35	ackdat	Float
11	dttl	Integer	36	is_sm_ips_ports	Binary
12	sloss	Integer	37	ct_state_ttl	Integer
13	dloss	Integer	38	ct_flw_http_mthd	Integer
14	service	nominal	39	is_ftp_login	Binary
15	Sload	Float	40	ct_ftp_cmd	integer
16	Dload	Float	41	ct_srv_src	integer
17	Spkts	integer	42	ct_srv_dst	integer
18	Dpkts	integer	43	ct_dst_ltm	integer
19	swin	integer	44	ct_src_ltm	integer
20	dwin	integer	45	ct_src_dport_ltm	integer
21	stcpb	integer	46	ct_dst_sport_ltm	integer
22	dtcpb	integer	47	ct_dst_src_ltm	integer
23	smeansz	integer	48	attack_cat	nominal
24	dmeansz	integer	49	Label	binary
25	trans_depth	integer			

Table 4
The UNSW-NB15 dataset attack categories.

Traffic	Description
Normal	The traffic is completely safe.
Fuzzing	An automated software testing technique aims to detect software flaws.
Analysis	Analysis system input/output to gain something about the target.
Backdoor	A malware family is used to gain illegal remote access to a computer.
DOS	Illegal authentication requests overwhelm the network/server.
Exploits	Code that takes advantage of a flaw or weakness in software.
Generic	A “generic attack” on a cryptographic primitive can be carried out, regardless of the specifics of how the cryptographic primitive is implemented.
Reconnaissance	Information gathering techniques aim to collect information about a network or server.
Shellcode	An attack is made up of instructions introduced and executed by a flawed program.
Worms	Self-replicating malicious code consumes the network resource and reduces its availability.

each step of this phase significantly impacts the model performance.

3.2.1. Feature encoding

ML-based models necessitate numeric input and output variables.

¹ <https://www.kaggle.com/datasets/mrwellsdavid/unswnb15>

This means that you must encode categorical data into numbers before fitting and evaluating a model when found within that data. As stated in 3.1, the security features that must be encoded are proto, service, state, and attack_cat. The two most prevalent techniques are a LabelEncoder and One Hot encoding [53]. Because the One Hot Encoding method can significantly increase feature dimensions by adding new features based on the unique values in each nominal feature [54], we decided to use Label Encoding to encode all of the nominal security features in the dataset. To this end, python *sklearn* is used to implement *LabelEncoder*.

3.2.2. Feature scaling

Feature scaling is a method for normalizing independent variables or data features. This can be achieved using two widely used methods - Normalization (Min-Max scaler) and Standardization. Normalization is a scaling technique that rescales features to fall within the [0,1] range. Standardization is a scaling technique that rescales features to give them the properties of a standard distribution with a mean of 0 and a standard deviation of 1. The standardization algorithm is determined as shown in (1).

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

Where Z is the new scaled value, μ is the distribution means, and σ is the standard deviation of the distribution.

3.2.3. Features selection

Selecting the most appropriate attributes is critical when spotting attack behavior. Eliminating superfluous features enhances an IDS's overall efficiency while increasing detection accuracy and speeding up computation. Therefore, feature selection (FS) and ranking are essential for ML-IDS models [55]. The Gini index (GI) and information gain are two common feature selection approaches. The GI is simple to implement for Classification and Regression Trees (CART), especially for high-dimensional features [56]. CART is a decision tree learning algorithm that recursively partitions data based on the feature with the highest GI score [CART stands for Classification and Regression Trees]. The GI is an FS method that measures the purity of features in a given class. A feature's purity refers to how well it can distinguish between distinct groups. If we consider dataset D, which includes samples from k different groups and P_i Denotes the probability of samples linked to class i at a given node, So the Gini impurity of the dataset D would be as described according to [57,58] (2) as follows:

$$G(D) = 1 - \sum_{i=1}^k (P_i)^2 \quad (2)$$

We used Python and the ITMO FS (Information Technologies, Mechanics and Optics University FS) library [59] to calculate the Gini index (GI) for all security features. The ITMO FS library provides a better view of traditional feature selection algorithms, supports hybrids and ensembles, and is more user-friendly than other libraries. Furthermore, our tests showed that the ITMO FS library works more efficiently than the largest Python feature selection ASU library on some algorithms. After extracting the top 13 impurity features (Fig. 2) based on a selection threshold of 0.15, we constructed an effective tree-based intrusion detection model. This threshold represents the maximum impurity of a feature to be included in the model. By selecting a threshold of 0.15, we were able to pick the top 13 features that had an impurity score below 0.15. This threshold was determined based on experiments, where we followed the best accuracy approach with a minimum feature space.

4. Experiments and discussion

The experiments were carried out on a laptop with an Intel Core i5-5300 U processor running at 2.30 GHz, 8 GB of RAM, and a 64-bit Windows 10 operating system. The experiments are written in the Python programming language.

Table 5 shows the experimental implementation environment.

4.1. Performance evaluation metrics

We compared our model with other baseline approaches to evaluate the proposed intelligent tree-based DDoS detection model. The confusion matrix defines our performance measurements, including Accuracy (3), Recall (4), Precision (5), and Fscore (6). In addition, the confusion matrix summarises classification problem prediction outcomes, and the number of correct and unsuccessful predictions is totaled and broken down by class using count values. The following derivations are derived

Table 5
Implementation environment.

Element	Description
Personal Computer	Windows 10, 2.30 GHz processor speed and 8 GB RAM
Programming Language	Python version 3.8
Dataset Management	Delimit 3.7.

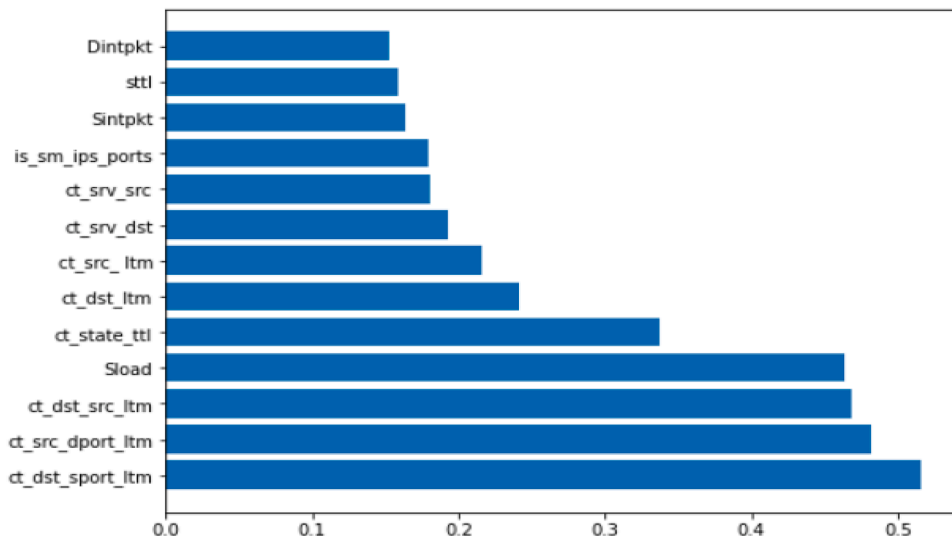


Fig. 2. Selected features based on a 0.15 threshold.

from the confusion matrix: True Positive (TP), False Positive (FP), and True Negative (TN).

$$Accuracy = \frac{(TP + TN)}{TP + TN + FP + FN} \quad (3)$$

$$Recall(Sensitivity) = \frac{TP}{(TP + FN)} \quad (4)$$

$$Precision = \frac{TP}{(TP + FP)} \quad (5)$$

$$Fscore = \frac{Precision \times Recall}{Precision + Recall} \quad (6)$$

4.2. Results and comparison

In this section, we compare our proposed intelligent DDoS attack detection model based on the DT and Gini Index (GI) FS method with the existing works in the literature. Our proposed model is based on the UNSW-NB15 dataset, which is a more recent and comprehensive dataset than those used in previous works. In addition, our proposed model uses an enhanced Gini index feature selection approach to reduce the data dimensionality and avoid overfitting issues, which is a significant improvement compared to the previous models. The enhanced Gini index approach allowed us to select the top 13 most important security features for constructing our effective tree-based intrusion detection model. After calculating the Gini index for all security features using the ITMO FS library, we selected the top 13 most important features based on a selection threshold of 0.15. This threshold was chosen based on experiments using an approach that maximized accuracy while minimizing feature space. By selecting the top 13 features, we constructed an effective tree-based intrusion detection model with fewer features, making it computationally efficient and less prone to overfitting. Additionally, we observed that these 13 features were sufficient to achieve a high level of accuracy in detecting DDoS attacks, indicating that they were indeed the most important features for this particular use case.

Our proposed model was developed and evaluated on the UNSW-NB15 dataset, a more recent and comprehensive dataset than those used in previous works. In addition, the dataset includes 1140,045 samples, which is larger than the datasets used in many previous studies. This larger dataset provides more reliable and realistic results and ensures that dataset limitations do not affect the proposed model's performance.

Our proposed model achieved an accuracy, Precision, Recall, and F-score of 98%, as shown in Table 6. These results indicate that our model can accurately detect attacks while minimizing false positives and false negatives. The high accuracy and other performance metrics demonstrate the effectiveness of our proposed approach. In addition, our proposed model uses an enhanced Gini index feature selection approach to reduce the data dimensionality and avoid overfitting issues. This approach allowed us to select the top 13 security features for constructing our effective tree-based intrusion detection model. The feature selection step was important to reduce the model's complexity and enhance its performance.

The proposed model's classification report includes Precision, Recall, F1-score, and support metrics for all attack categories in the UNSW-NB15 dataset is illustrated in Table 7. Precision is the proportion of true positives to the total number of predicted positives. The Recall is the proportion of true positives to the total number of actual positives. The F1-score is the weighted harmonic mean of Precision and Recall. Finally,

Table 7

The proposed model's classification report in its entirety.

Attack	Precision (%)	Recall (%)	F1-score (%)	Support
Analysis	5.8	13.2	8.0	159.0
Backdoor	4.6	17.3	7.3	98.0
DoS	34.1	29.0	31.4	2217.0
Exploits	67.0	59.2	62.9	5670.0
Fuzzers	56.4	59.6	57.9	2945.0
Generic	98.1	99.0	98.6	20,600.0
Normal	99.7	99.7	99.7	308,715.0
Reconnaissance	79.2	88.0	83.4	1412.0
Shellcode	40.4	42.7	41.5	178.0
Worms	30.4	35.0	32.6	20.0
Accuracy	98.0	98.0	98.0	1.0
Macro avg	51.6	54.3	52.3	342,014.0
Weighted avg	98.0	98.0	98.0	342,014.0

the support metric refers to the number of instances of a particular attack category in the dataset.

Looking at the precision metric, the proposed model achieved excellent performance in identifying generic attacks with 98.1%, while it has a very low precision score in recognizing Backdoor (4.6%) and Analysis (5.8%) attacks. A low precision score indicates that the proposed model has a high false-positive rate in identifying these attack categories. The proposed model's precision scores in identifying Dos, Exploits, Fuzzers, Normal, and Reconnaissance attacks were 34.1%, 67.0%, 56.4%, 99.7%, and 79.2%, respectively.

The Recall metric shows that the proposed model performed well in identifying Normal attacks (99.7%). However, the recall score for other attacks was relatively lower, with the model correctly identifying only 13.2% of Analysis attacks and 17.3% of Backdoor attacks. The recall score for the remaining attacks, including DoS, Exploits, Fuzzers, Generic, Reconnaissance, Shellcode, and Worms, ranged from 29.0% to 88.0%.

The F1-score metric measures the proposed model's overall accuracy in identifying attacks. The proposed model achieved the highest F1-score in recognizing Normal attacks (99.7%). The F1-score for the remaining attacks ranged from 7.3% to 83.4%.

Finally, the support metric indicates the number of instances of each attack category in the dataset. The proposed model correctly identified 308,715 normal traffic instances, which is the largest category in the UNSW-NB15 dataset. In contrast, the smallest category in the dataset, Worms, was only represented by 20 instances. The support metric enables us to interpret the results of the other metrics in the context of the number of instances of each attack category.

The detailed performance metrics output comparison for each baseline approach proposed by [43] is shown in Table 8. The proposed model outperforms the others in terms of all performance metrics. Furthermore, by focusing on the most important security features, we reduced the computational complexity in time and over-fitting (reduced number of security features).

A confusion matrix is a vital tool in evaluating the performance of classification models. It illustrates how well the model has predicted the true labels and how often it has misclassified them. The confusion matrix in Fig. 4 Confusion matrix. Shows the performance of the proposed intelligent DDoS attack detection system based on the DT and Gini Index (GI) FS method in classifying the different types of attacks in the UNSW-NB15 dataset. The sample data size of 1140,045 was split into 70% for training and 30% for testing. The diagonal elements in the confusion matrix represent the number of instances that the proposed model for each class correctly classified. For example, the model correctly classified all 308,715 "Normal" class instances, corresponding to a true positive rate of 100%. The model also correctly classified 98.6% of the instances in the "Generic" class, which is the most common class in the dataset. The off-diagonal elements in the confusion matrix represent the instances that the proposed model misclassified. For example, the model misclassified 15 instances of the "Analysis" class as "Backdoor" and 35

Table 6
Result of experiments.

Accuracy	Precision Recall	F-score
98%	98% 98%	98%

Table 8

Performance of the proposed model in comparison to other approaches.

	Algorithm	Precision	Recall	Fscore	Dataset Name	Size	Feature
Proposed Work	DT	98%	98%	98%	UNSW-NB15	1140,045	13/45
Ismail et al. (2022) [43]	RF	89%	89%	89%	UNSW-NB15	82,332	45/45
Ismail et al. (2022) [43]	XGBoost	90%	90%	90%	UNSW-NB15	82,332	45/45

instances of the “Fuzzers” class as “Exploits”. The confusion matrix also shows that the model had some difficulty in correctly classifying the “DoS”, “Reconnaissance”, and “Shellcode” classes, with true positive rates of 31.4%, 83.4%, and 41.5%, respectively.

The confusion matrix indicates that the proposed model has a high accuracy and a low false alarm rate. Out of the approximately 34,000 testing instances, only 6800 examples were misclassified, which is an error rate of 2%. This suggests that the proposed model can effectively detect a wide range of attacks in network traffic data with high accuracy.

Compared to previous works in the literature, our proposed intelligent DDoS attack detection system based on DT and Gini index feature selection achieved remarkable performance on the UNSW-NB15 dataset, which contains 1,140,045 samples. To further evaluate the strength of our approach, we compare its performance with the baseline work in [40], which also used the UNSW-NB15 dataset to develop two intrusion detection models based on Random Forest and XGBoost for DDoS attacks. In this section, we present a detailed comparison between our proposed model and the baseline work, highlighting the strengths and advantages of our approach. Table 8 compares the performance of our proposed model with that of two other approaches. Ismail et al. [43] also used the UNSW-NB15 dataset to develop two models for DDoS attack detection, one based on the Random Forest algorithm and the other based on XGBoost. The proposed model achieved an overall accuracy of 98%, with both precision and recall values of 98%. In comparison, the Random Forest and XGBoost models developed by Ismail et al. achieved accuracy scores of 89% and 90%, respectively, with both models having a larger feature space of 45 features.

It is worth noting that the UNSW-NB15 dataset used in our proposed model contains 1,140,045 samples, significantly larger than the dataset size used in the baseline paper by Ismail et al. [43] had a dataset size of 82,332 samples. This larger dataset size makes our model more robust and accurate in detecting DDoS attacks.

Our proposed model’s performance on the UNSW-NB15 dataset is particularly noteworthy because it achieved a high level of accuracy

using only 13 out of the original 45 features. This result indicates that the proposed feature selection approach based on the Gini index effectively reduces the dataset’s dimensionality while maintaining high accuracy levels.

Moreover, Fig. 3 presents a visual comparison of the accuracy of the proposed model with that of the Random Forest and XGBoost models from the baseline paper. As shown in Fig. 3, our proposed model outperforms the baseline models in terms of accuracy, with an overall accuracy score of 98% compared to 89% and 90% for the Random Forest and XGBoost models, respectively.

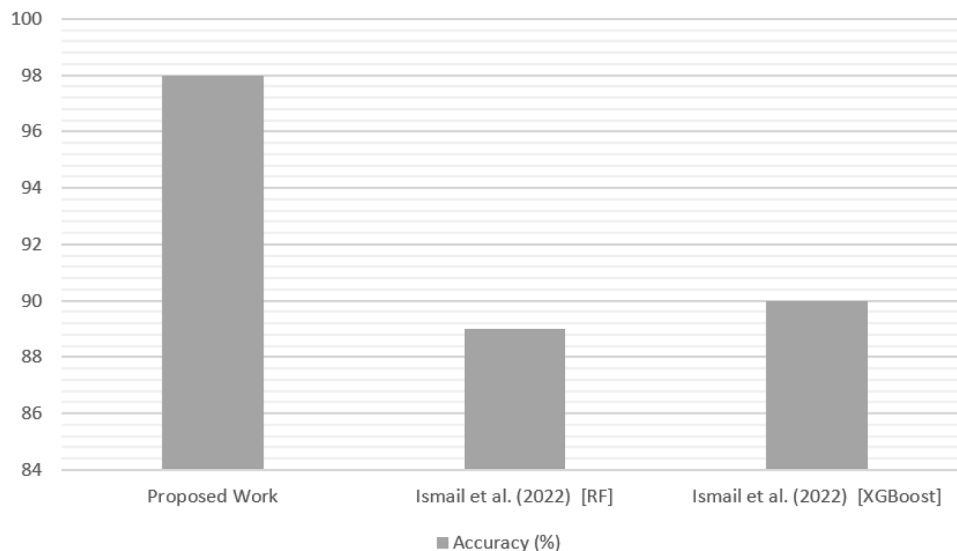
Overall, the performance of our proposed model on the UNSW-NB15 dataset indicates its novelty and strength in accurately detecting DDoS attacks while using a reduced feature space. In addition, the use of a more significant dataset size and an enhanced feature selection approach based on the Gini index also contribute to the proposed model’s high accuracy and robustness.

5. Conclusion

In this work, we proposed a intelligent DDoS attack detection system based on a DT algorithm and Gini index feature selection. Our proposed model was trained and tested on the UNSW-NB15 dataset, which contains over 1 million network traffic samples with various types of attacks. By selecting only the 13 most important security features and applying our enhanced Gini index approach, we achieved an overall accuracy of 98%, outperforming previous approaches that used more complex models and more features.

Our model demonstrated high precision and recall rates for all types of attacks, with only a small number of misclassifications. In addition, the confusion matrix analysis revealed that our model had a low false alarm rate, indicating that it was able to accurately detect and classify DDoS attacks while minimizing the number of false positives.

The comparison with the baseline work showed the superiority of our proposed model in terms of accuracy, feature selection, and dataset

**Fig. 3.** Visualizing the Accuracy of the Proposed Model against other Approaches.

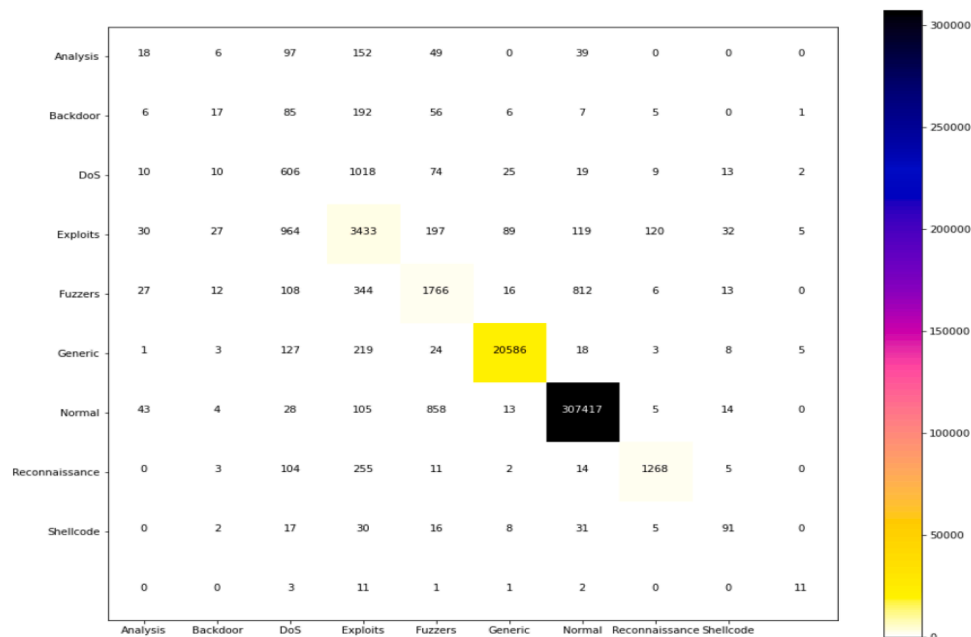


Fig. 4. Confusion matrix.

size. In addition, our model demonstrated strong novelty and practicality for real-world DDoS attack detection, especially in resource-constrained environments where efficiency and effectiveness are critical.

In conclusion, our proposed intelligent DDoS attack detection system based on a DT algorithm and Gini index feature selection is an effective and efficient solution for network security that can be used to detect DDoS attacks with high accuracy while minimizing false positives. This work has important implications for improving the security and reliability of networks and can be extended to other domains of network intrusion detection.

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Data availability

The data download source has been cited inside the paper.

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